Final Visualisation Report

Title: E-commerce Sales Analysis - A Journey Through Trends and Insights
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Introduction:

In this cut-throat age of the rapidly changing world of online trading, one key area of competition today is linked sales analysis across varied customer behavioral patterns-to put things in a nutshell; type of data available is potential and movement regarding the sales of various products categorized in customer sections over time. Meaningful trends and relationships present themselves, and questions surrounding them such as: When would the sales peak times be? Which product categories deliver the best results? How far does the effort of marketing equals result? Let's jump in and find out what this data tells us.

Getting to Know the Data:

The data file has a record count of 1,000, each representing categories such as product category, price, discount, marketing spends, and the number of units sold. There are classifications for customers into segments Occasional, Premium, and Regular. It has quickly run through some of the more salient points derived from the data.

- An average product price of $505.24 and an average discount of 24.9% can be quoted.
- Marketing spends around $4,912 on an average campaign.
- On an average, 30 units per transaction are sold.

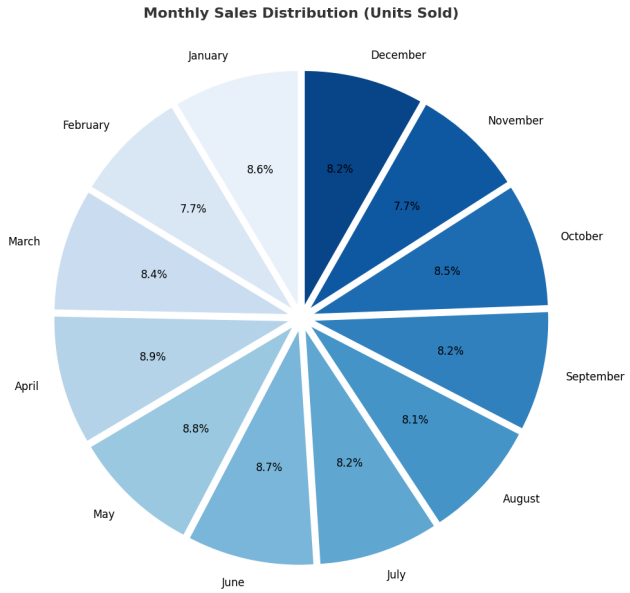
index	Mean	Median	Std Dev	Min	Max	Sum
Price	505.24192	509.31	289.7912481197464	14.59	999.42	505241.92
Discount	24.91975	24.965	14.43995129600273	0.01	49.92	24919.75
Marketing_Spend	4912.830239999999	4837.245000000001	2840.6026561655463	100.3	9995.62	4912830.239999999
Units_Sold	29.631	30.0	7.267988900835132	5.0	57.0	29631.0

What Do the Trends Show?

Seasonal Sales Patterns:

Time of Year Selling Patterns From the monthly sales distribution, it appears that April and May are also good months for generating almost equal 8.8% sales partly due to the typical buying-and-selling pattern or promotion.

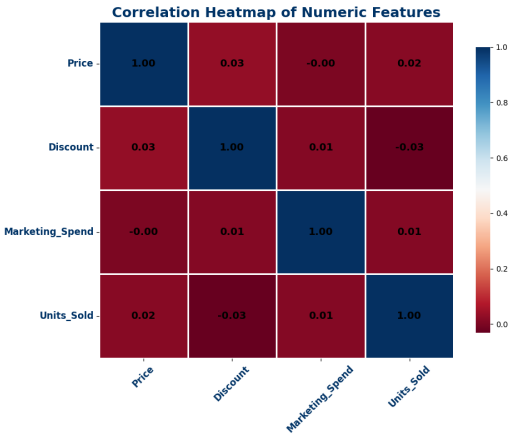
Conversely, this again interestingly brings sales figures down to around 7.7% in October and November. So, this merits the question: Are there chances or possibilities of increased sales during the otherwise quiet months? These patterns are clearly shown in the pie chart and easily point to the areas that need improvement.

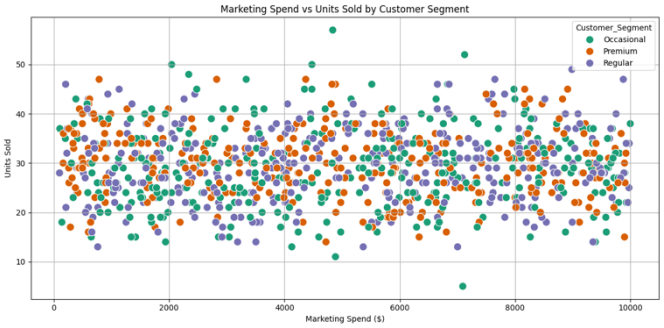


Exploring Relationships in the Data:

What Drives Sales?

Taking a closer look at the heatmap of correlation reveals a strange fact-an almost zero direct correlation exists between the amount spent on marketing and the number of units sold. It indicates that spending high on marketing does not guarantee a good number of sales. However, customer segmentation provides more insights: the scatterplot asking for Marketing Spend as opposed to Units Sold shows that indeed Premium customers yield more sales per Marketing dollar spent. Targeting these Premium customers with more personal campaigns and ways of converting these Occasional customers to Regular or Premium buyers.





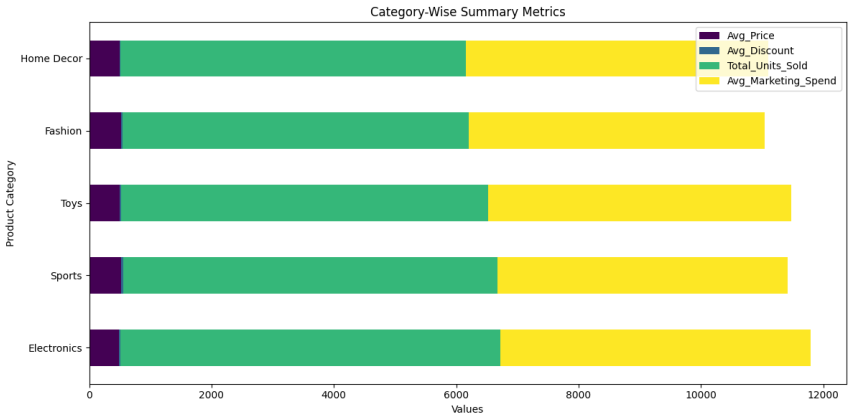
Which Categories Steal the Spotlight?

Top Performers:

Among product categories, Electronics and Fashion top the lists and contribute a lion's share to revenues. Such categories have all the ingredients for success-say, competitive pricing, organized marketing, and a good selection. Home Décor, on the other hand, happens to have potential. Most of the consumer price points tend to be higher instead of being lower-motivated against sales volume. This creates an opportunity for customer acquisition through better discounts or promotions targeted at these products.

Balancing the Mix:

The horizontal bar graph gives a clear indication of how different categories score on an average price, market spend, and total units sold. For example, toys and sports show relatively steady performance, but a more targeted marketing effort would help tap their growth potential.



Conclusion:

An insightful analysis of e-commerce reveals how data can be interpreted for actionable insights. Through the seasonal trends, understanding how customers behave, and viewing performance categories, businesses can effectively improve their strategies, allowing them to optimize their profits while increasing customer satisfaction. The very clear story from this data is that, with best focus and planning, there is room to grow and thrive in e-commerce.